



A rapid, novel approach to rare disease and clinical genetic variant discovery using on-flowcell proximity sequencing and haplotype-resolved variant calling

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Financial Disclosure

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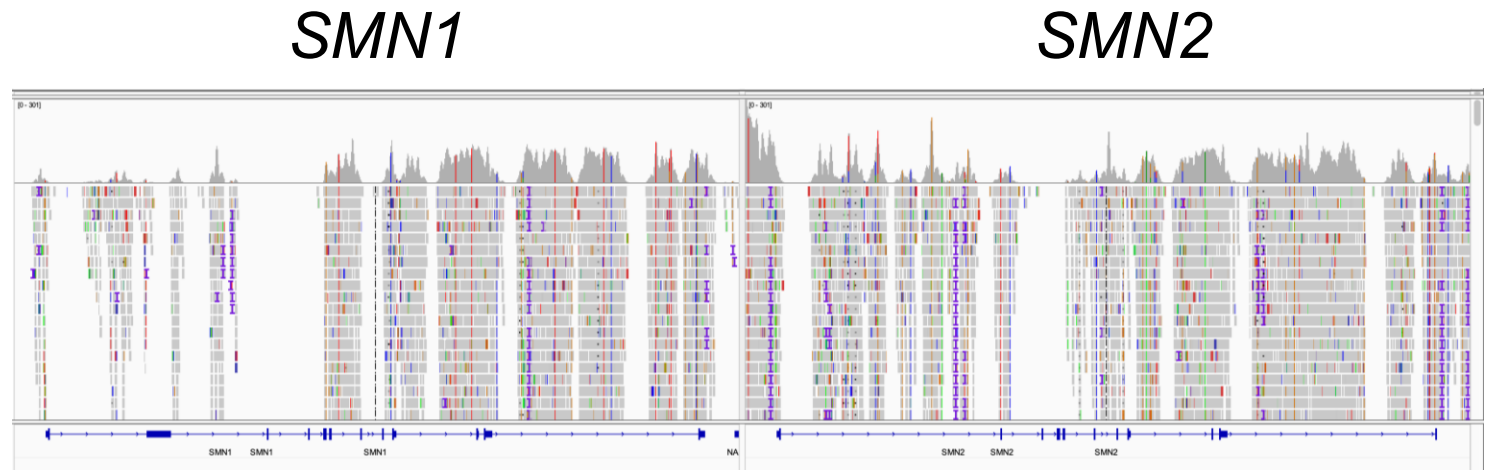
Discloses the following relevant relationships with ineligible companies. Any potential interests have been mitigated:

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Some medically relevant regions remain difficult for short-read sequencing (SRS)

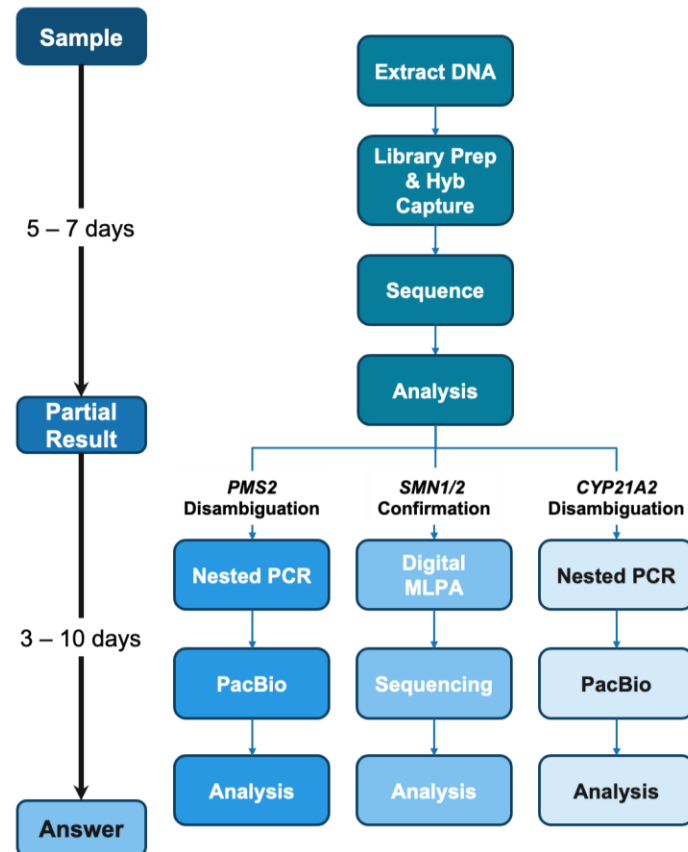
- 1168 genes have partial or complete sequence homology (> 98%) to one or more loci. Copy number and sequence variants in **193 of these 1168 genes are associated with rare Mendelian disorders**, inherited cancers and complex diseases (*Mandelker et al. 2016*).
- Disease-associated genes with mappability issues: ***SMN1/SMN2*** (Spinal Muscular Atrophy), ***PMS2*** (Lynch Syndrome), ***CYP21A2*** (Congenital Adrenal Hyperplasia), ***NCF1*** (Chronic Granulomatous Disease), ***STRC*** (Recessive Nonsyndromic Hearing Loss), they all have paralogs that share 98-99% homology.

Ambiguous or false read mapping due to high sequence homology in the reference



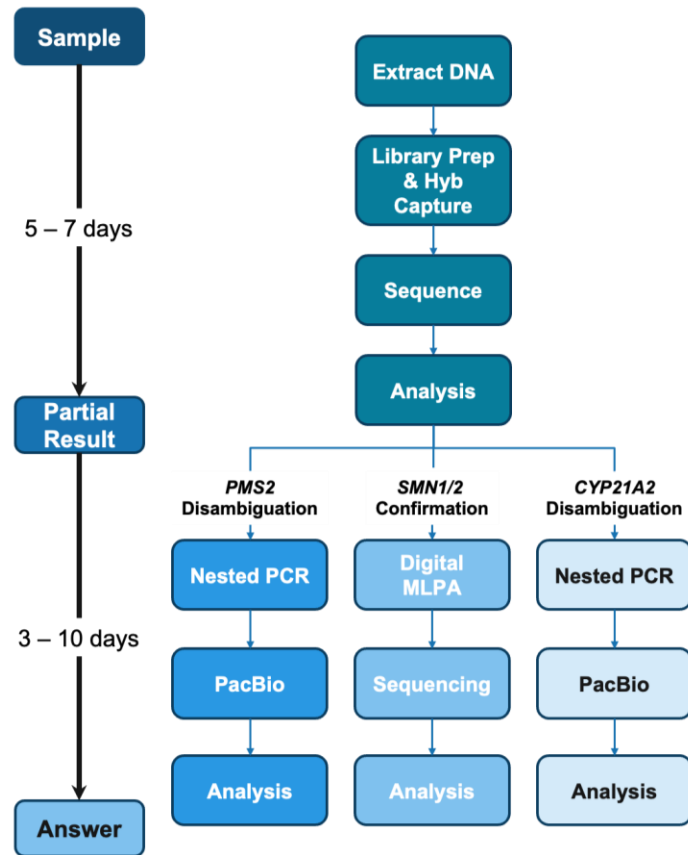
Current approaches to address these challenging genomic regions span > 1-2 weeks to generate final variant calls

Standard workflow

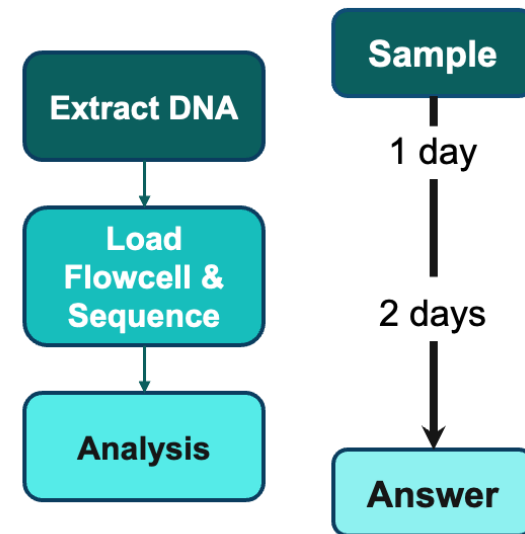


We present a simplified lab workflow, eliminating the need for secondary analytical methods to arrive at a final result

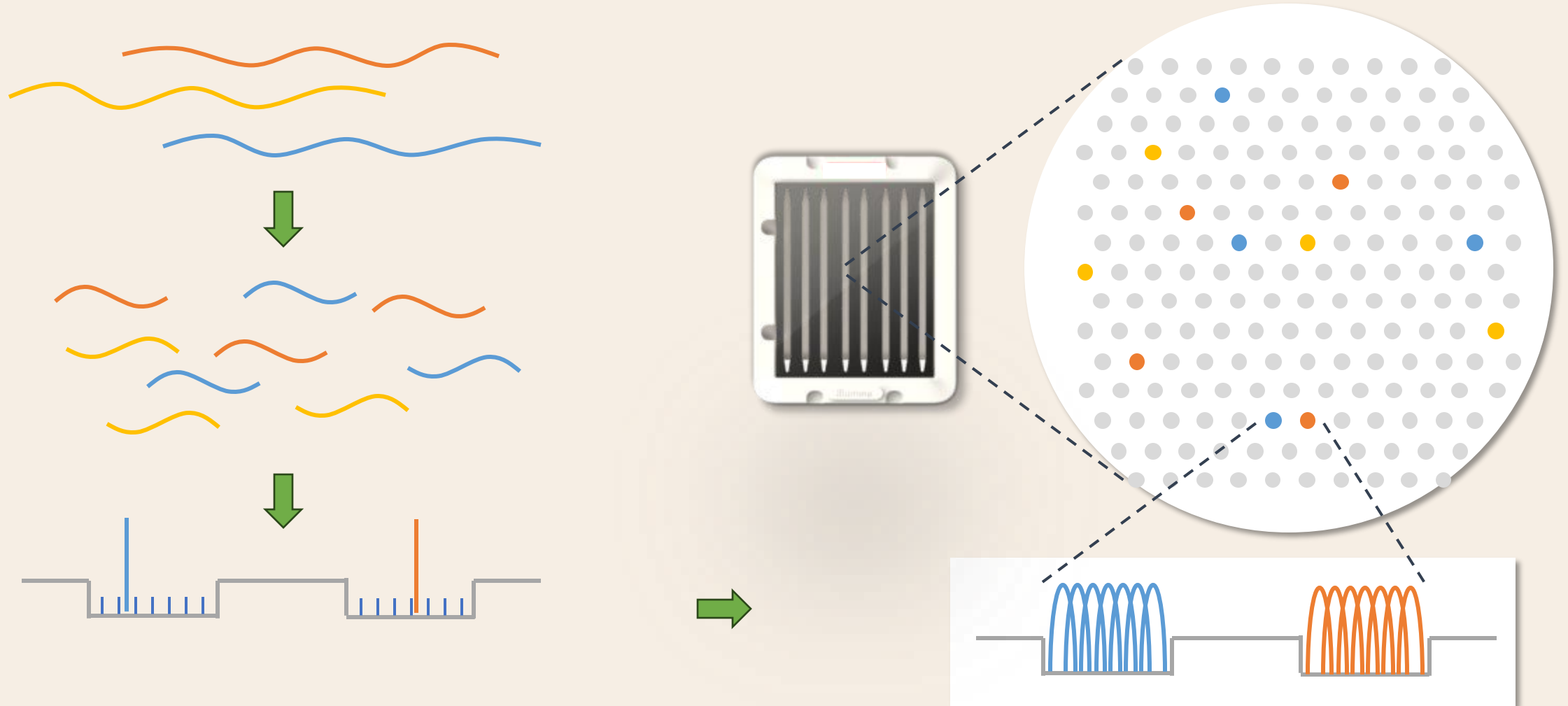
Standard workflow



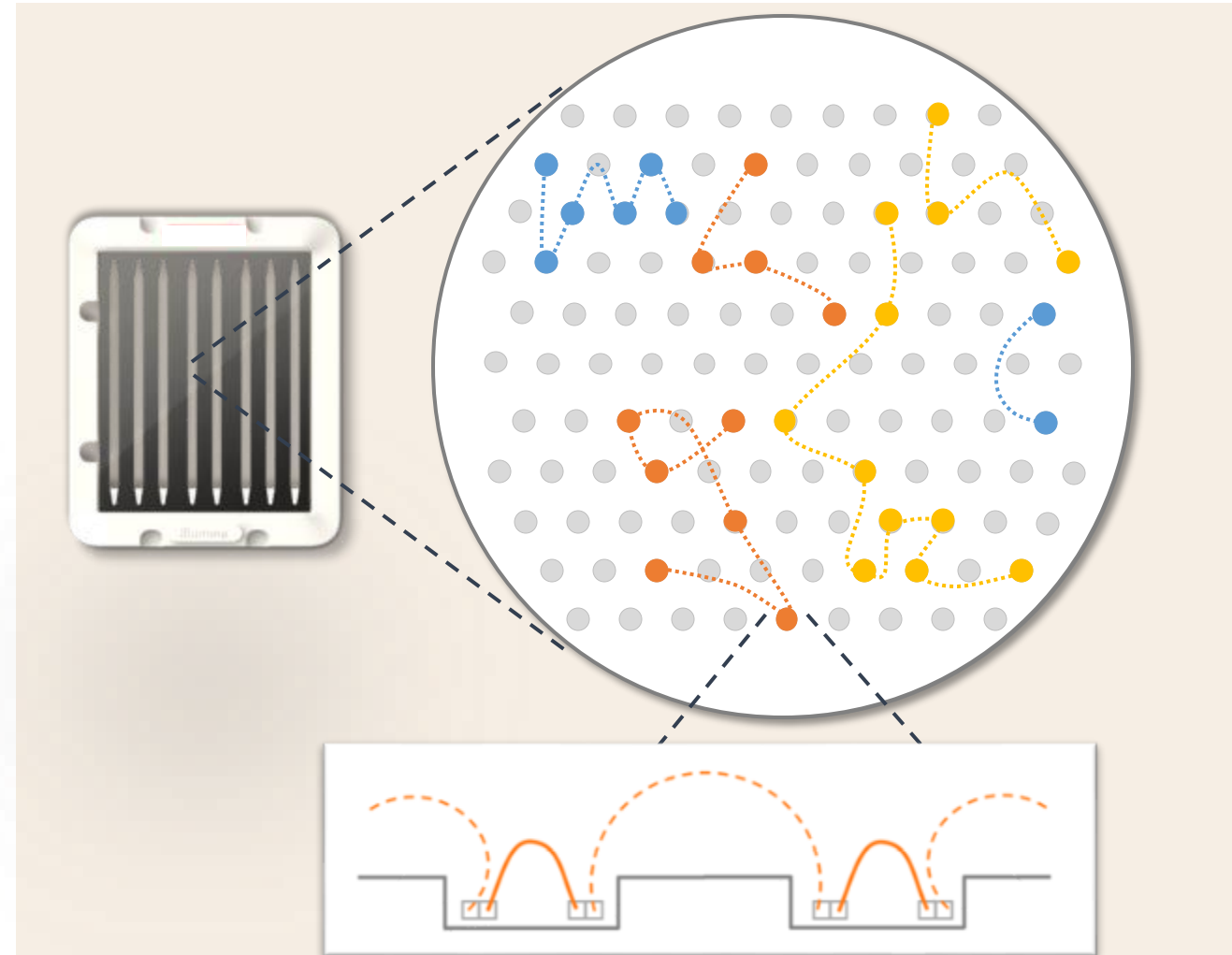
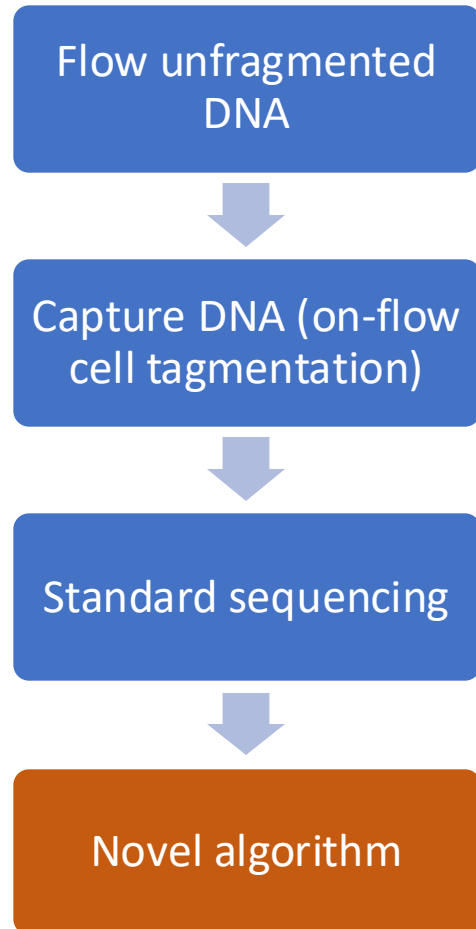
TruPath™ workflow



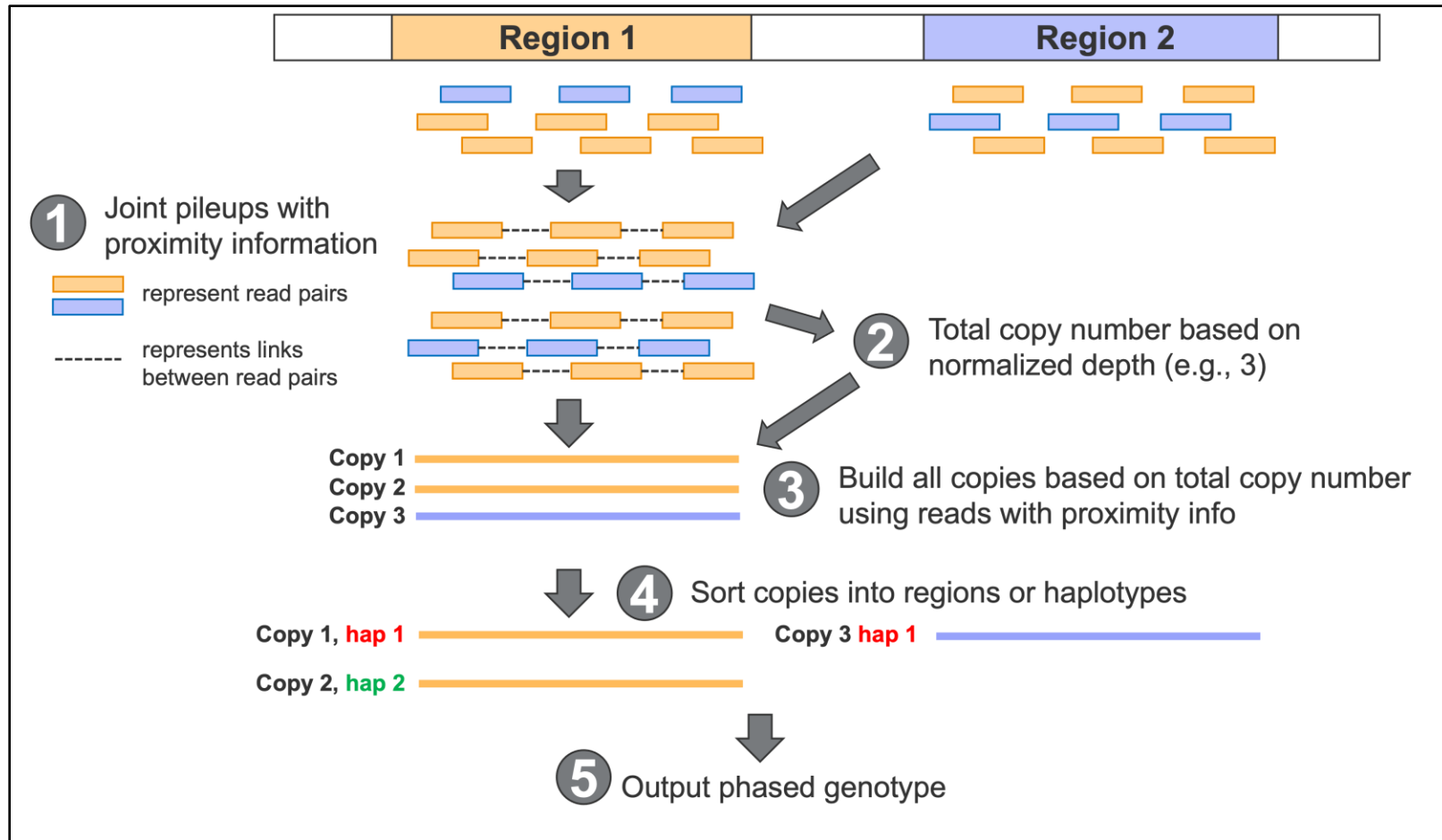
In standard SRS, DNA is first sheared into small, clusterable fragments, long-range linkage is lost



An on-flowcell proximity sequencing technology that enables rapid workflow and long-range phasing information



Multi-Region Joint Detection (MRJD) resolves paralogous regions by using proximity information from nearby nanowells encoded in the reads

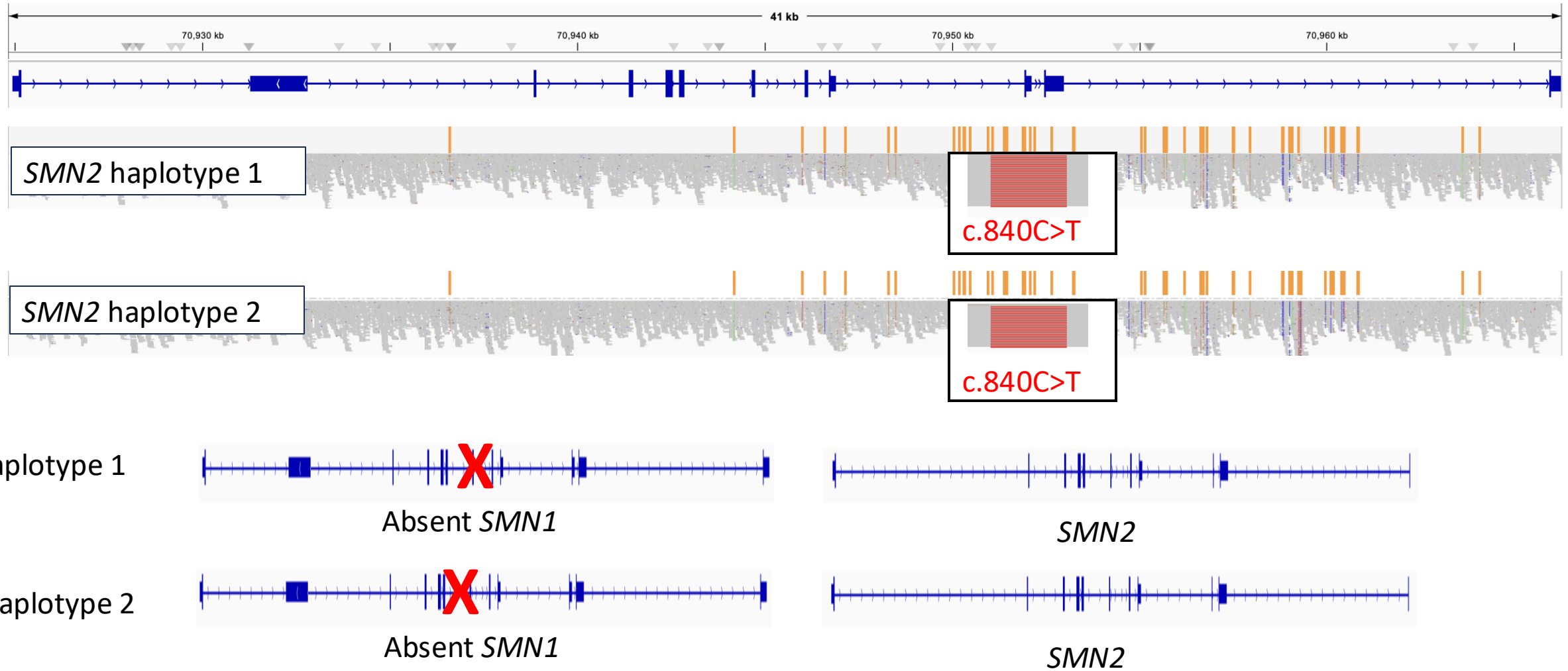


New workflow resolved known pathogenetic variants in 31/31 cases across eight paralogous genes

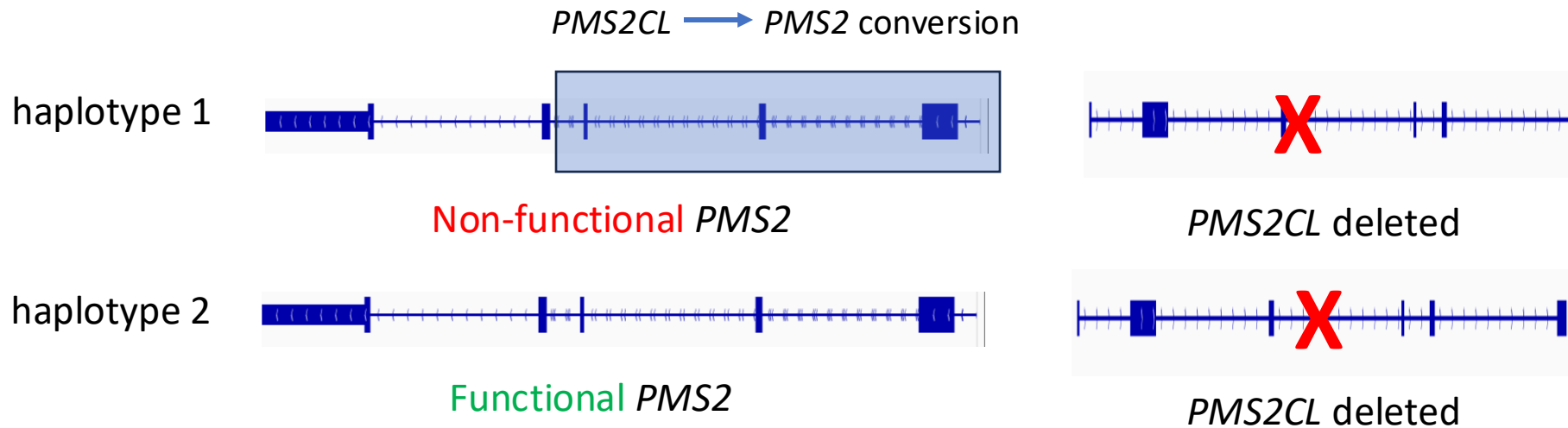
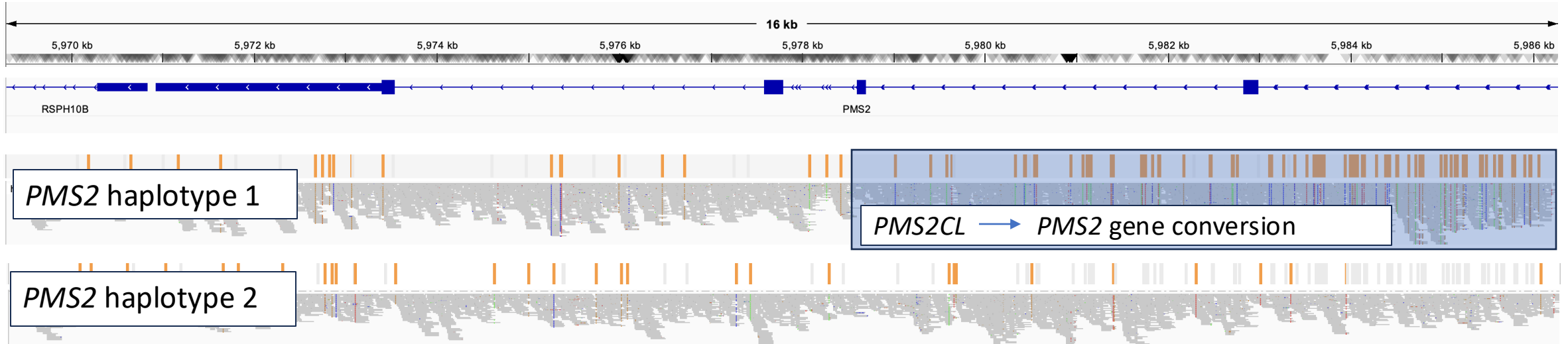
Gene	Variant types	Number of expected events observed with TruPath
<i>SMN1/SMN2</i>	SMN1 copy number loss	9/9
<i>PMS2</i>	Del/Dup, small variant	7/7
<i>CYP21A2</i>	Recombination, small variant	5/5
<i>STRC</i>	STRC copy number loss, small variant	3/3
<i>NCF1</i>	NCF1 copy number loss	3/3
<i>CFC1</i>	Small variant	1/1
<i>IKBKG</i>	Exon 4-10 del	2/2
<i>HYDIN</i>	Small variant	1/1

5 samples benefited from additional insights from TruPath

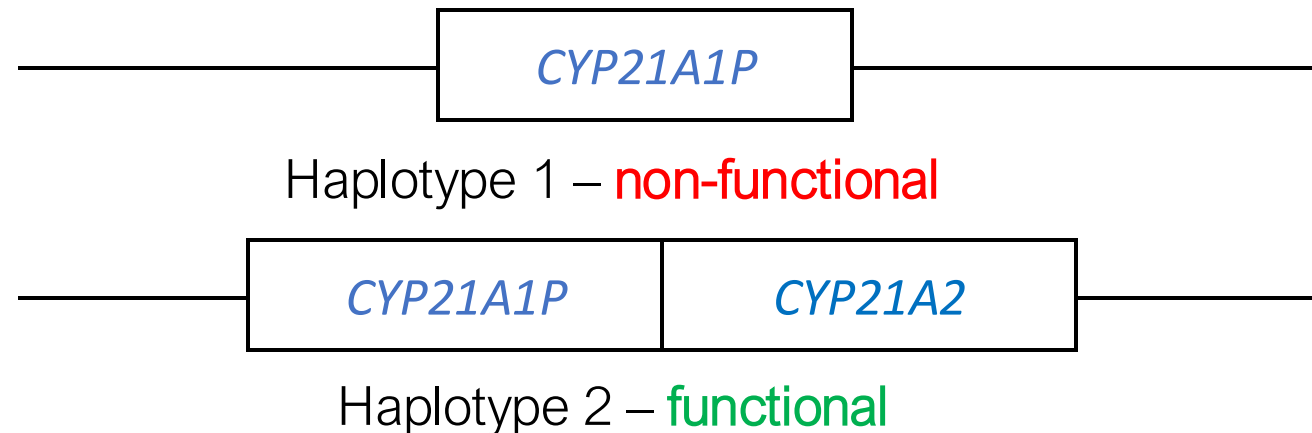
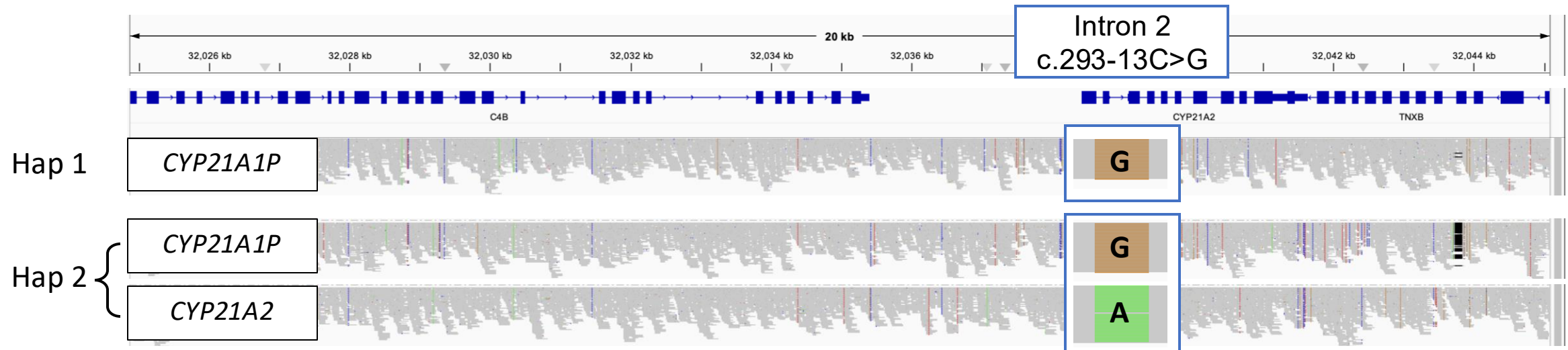
SMN1 and SMN2 Genotyping



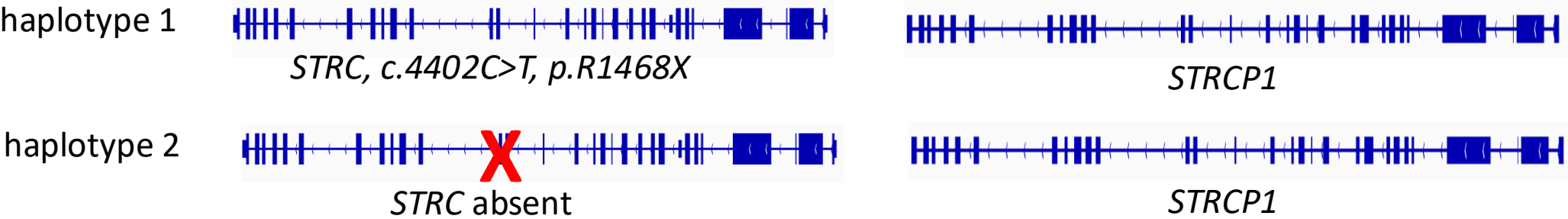
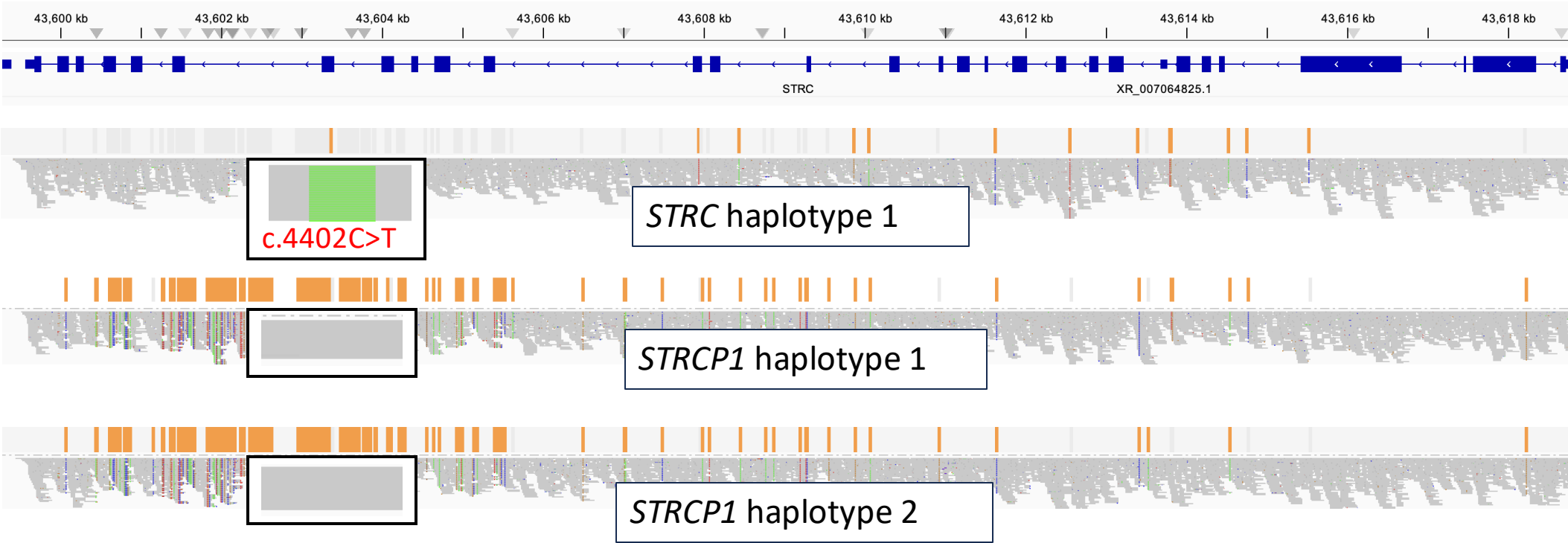
PMS2-PMS2CL ambiguity resolved



CYP21A2 locus structure resolved



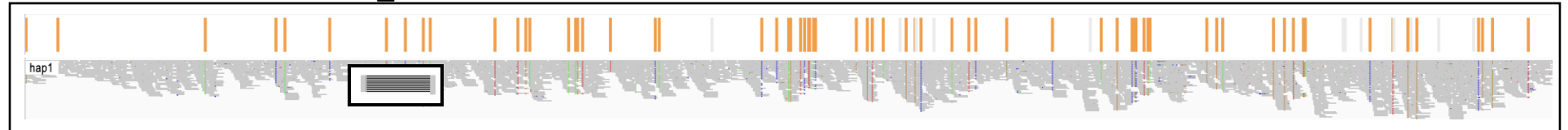
Biallelic disruption detected in *STRC*



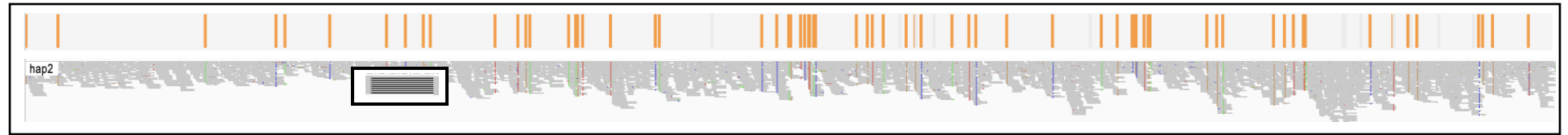
NCF1 loss due to putative gene conversion event

c.75_76del

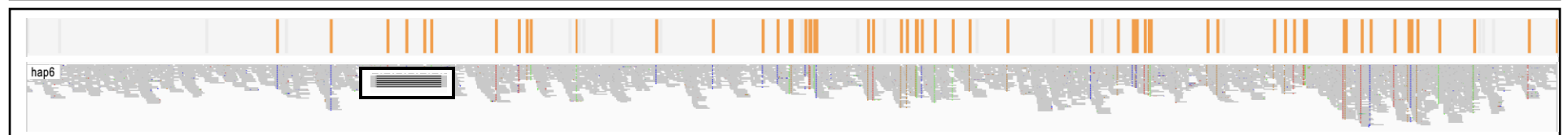
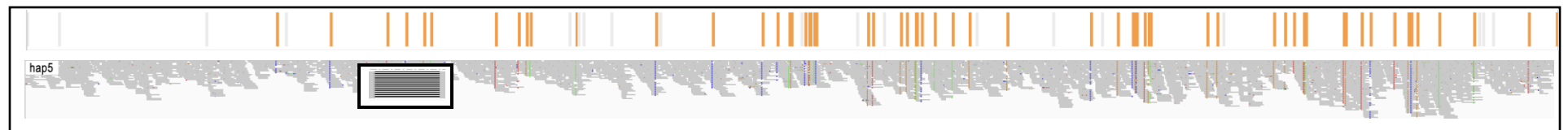
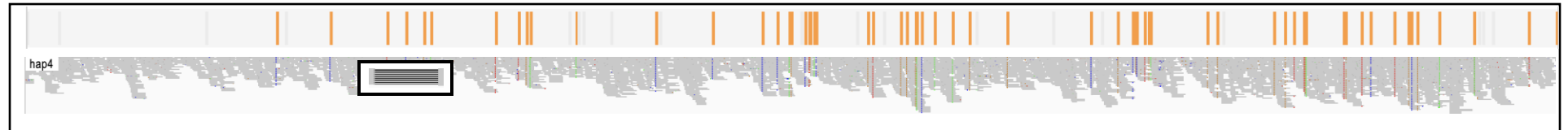
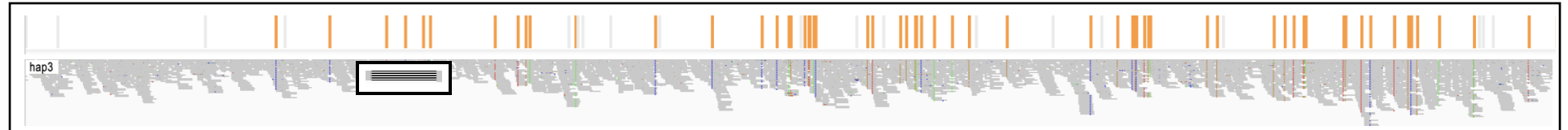
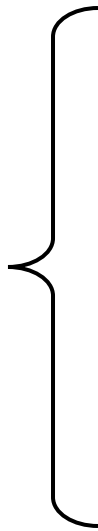
NCF1 haplotype 1



NCF1 haplotype 2



NCF1B/NCF1C
haplotypes



Summary

- TruPath resolved known pathogenetic variants in 31/31 cases across eight paralogous genes and produced phased, locus-wide haplotypes that clarified allelic configuration, gene-pseudogene assignment, and CNV context.
- More medically relevant and paralogous genes are expected to be included with this workflow.
- This data supports this approach as a rapid genome-wide assay that brings historically “dark” regions into routine clinical research while reducing the need for multiple orthogonal workflows with increased complexity, and time to result.

Acknowledgements

GeneDx, Labcorp, Broad Clinical Labs, UKSH

Illumina DRAGEN™ and TruPath team